SIMATS ENGINEERING

Engineer to Excel

SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab experiment-22

Experiment 22. Family Tree

Aim

To implement a family tree using Prolog.

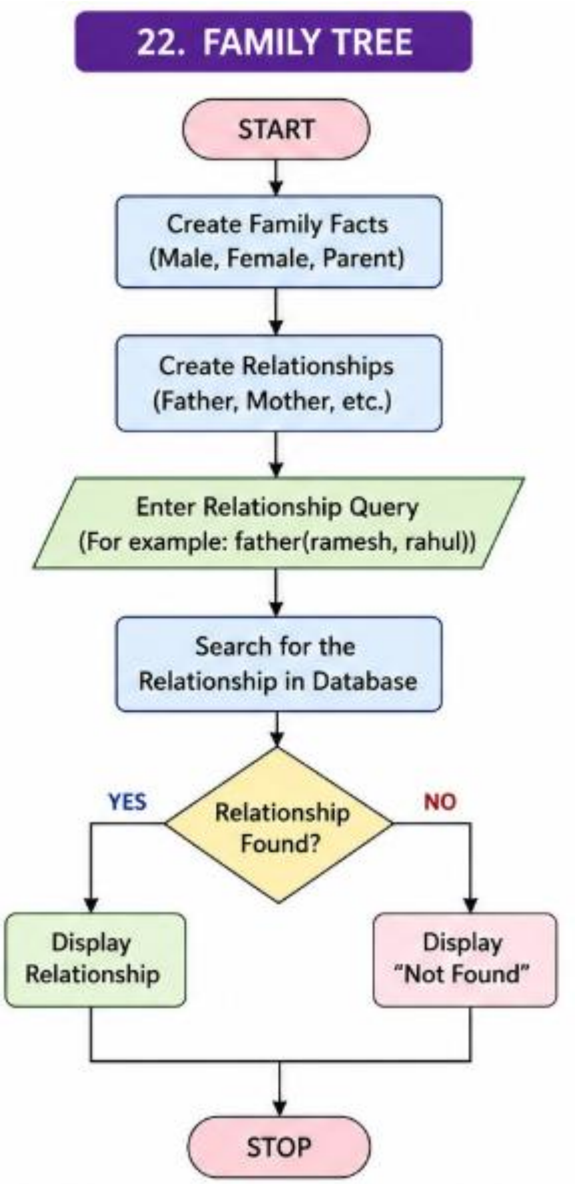
Objectives

- To represent family relationships.
- To understand Prolog facts and rules.
- To find relationships between family members.

Algorithm

1. Define male and female members.
2. Define parent relationships.
3. Create rules for father and mother.
4. Query the required relationship.
5. Display the result.

Flowchart



Code

```
male(ramesh).

male(rahul).
```

```
female(lakshmi).

female(priya).

parent(ramesh, rahul).

parent(lakshmi, rahul).

parent(ramesh, priya).

parent(lakshmi, priya).

father(X, Y) :-

    male(X),

    parent(X, Y).

mother(X, Y) :-

    female(X),

    parent(X, Y).

main :-

    father(ramesh, rahul),

    write('Ramesh is the
father of Rahul.').

:- initialization(main).
```

Output

HelloWorld.pl

```
male(ramesh).
male(rahul).

female(lakshmi).
female(priya).

parent(ramesh, rahul).
parent(lakshmi, rahul).
parent(ramesh, priya).
parent(lakshmi, priya).

father(X, Y) :-
    male(X),
        parent(X, Y).

        mother(X, Y) :-
            female(X),
                parent(X, Y).

                main :-
                    father(ramesh, rahul),
                        write('Ramesh is the father of Rahul.').

                        :- initialization(main).
```

I/O AI Agent

STDIN

Input for the program (Optional)

Output:

Ramesh is the father of Rahul.

Result

Ramesh is the father of Rahul.

Conclusion

Thus, the family tree was successfully implemented using Prolog.